Lab 3 – Challenge 2

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GitHub_lab_3

Problem/Question: How did population change across Bristol's wards between 2012 and 2022?

- Did some wards experience much greater increases than others?
- Did wards with larger starting populations in 2012 experience greater increases in population by 2022?

Data source: Bristol City Council Open Data

ArcGIS REST APIs used:

1) **Ward population profiles:** population estimates in Bristol wards including counts by age and gender for each year from 2012-2022
2) **Electoral Ward Boundaries:** polygon geometries for Bristol's 34 electoral wards

Chose to use APIs not a downloaded dataset to make workflow more easily reproducible.

Dataset:

1) Ward population profiles: initially returned 1,155 records with 97 columns - including the different ages, genders and years

Filtering process:

- Filtered GENDER column to All Persons count (per ward) and removed the Bristol-city summary record (would affect descriptive stats)
- Filtered to relevant columns: WARD_CODE, WARD_NAME, PERIOD and ALL_AGES
- Filtered to only two years to compare: 2012 &2022
- Calculated absolute population percentage change as proportional increase depends on starting population

Data Challenges:

1)Double-nested API response:

- Initially tried to convert to DataFrame per usual method but that returned individual population attributes within another dictionary not as separate columns
- Inspected JSON response closer and saw that the population values were nested inside 'features'->'attributes'
- Used for loop and .append() function (like the lab 2 Glasgow API example work through), to extract the attributes from the dictionary

2) Getting the ward geometry:

- Fist attempt to get the ward geometry I extracted only attributes and no ward geometry
- Checked API documentation and switched from JSON to GeoJSON and used gpd.read_file() to instead read the data directly into GeoPandas to keep the geometries

3) Joining population + ward boundary data:

- For ward identification, the ward population used WARD_CODE and the ward boundaries used WARD_ID
- So I used left_on and right_on for pd.merge() – and the resulting GeoDataFrame confirmed the matched 34 wards and all columns values were intact and matching.

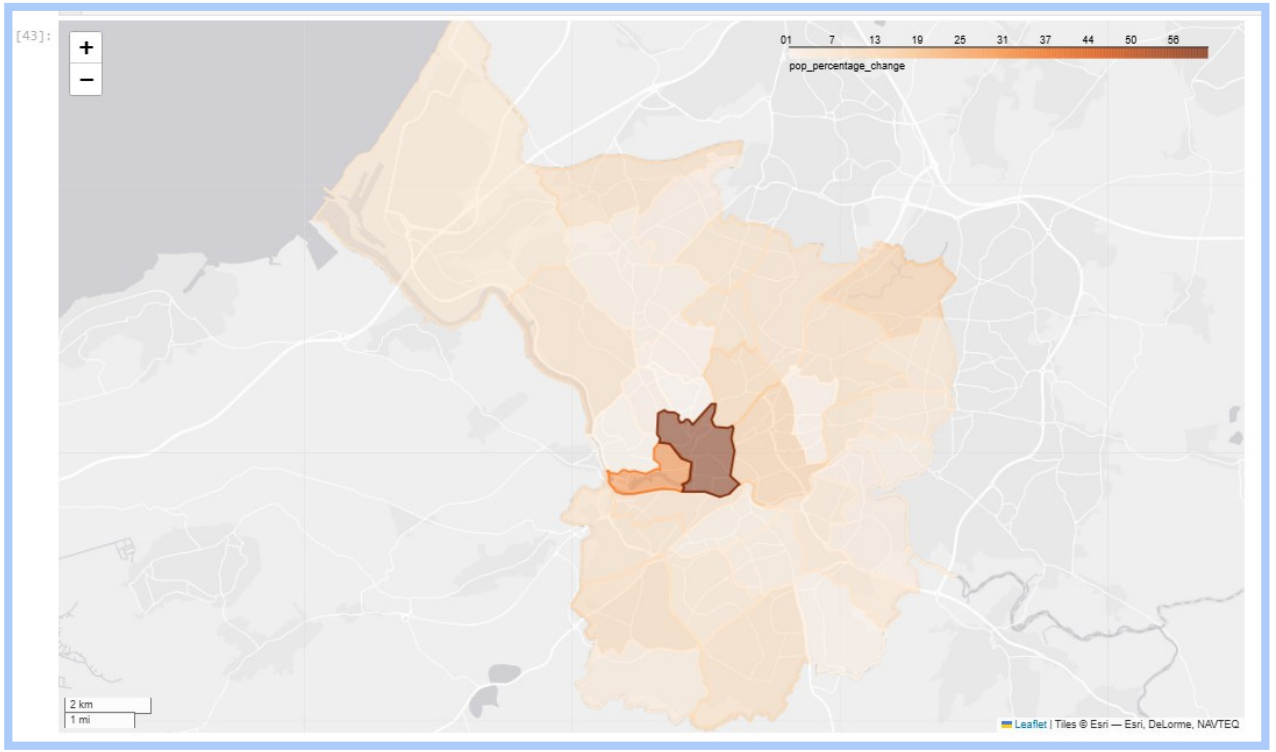


Figure 1: Population percentage change across Bristol wards 2012-2022

Used GeoPandas.explore() to create this interactive choropleth. I chose percentage rather than absolute population increase for the outcome to make comparisons between wards with different initial populations more meaningful.

Map highlights the variation seen in population growth spatially, with Central and Hotwells and Harbourside experiencing the most visual contrast.

Insights: Statistical Findings and interpretations

Descriptive statistics:

-After calculating population change for each ward, used Pandas .describe() function

- The average ward population increase was 10.81% but the median was 9.26% suggesting a few high growth wards are influencing mean value
- The relatively large standard deviation of 10.61 also indicates a variable growth across wards

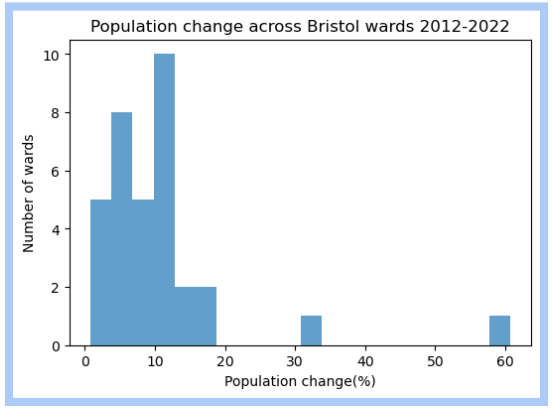


Figure A: Distribution of population change histogram

Pandas histogram function to show how the percentage population changes were distributed across the 34 wards.

Most wards experienced modest increases of around 0-15% - with distribution right skewed due to a few with higher increases in that cluster. Central shows a 60% increase and is an obvious outlier – explaining the median and SD values.

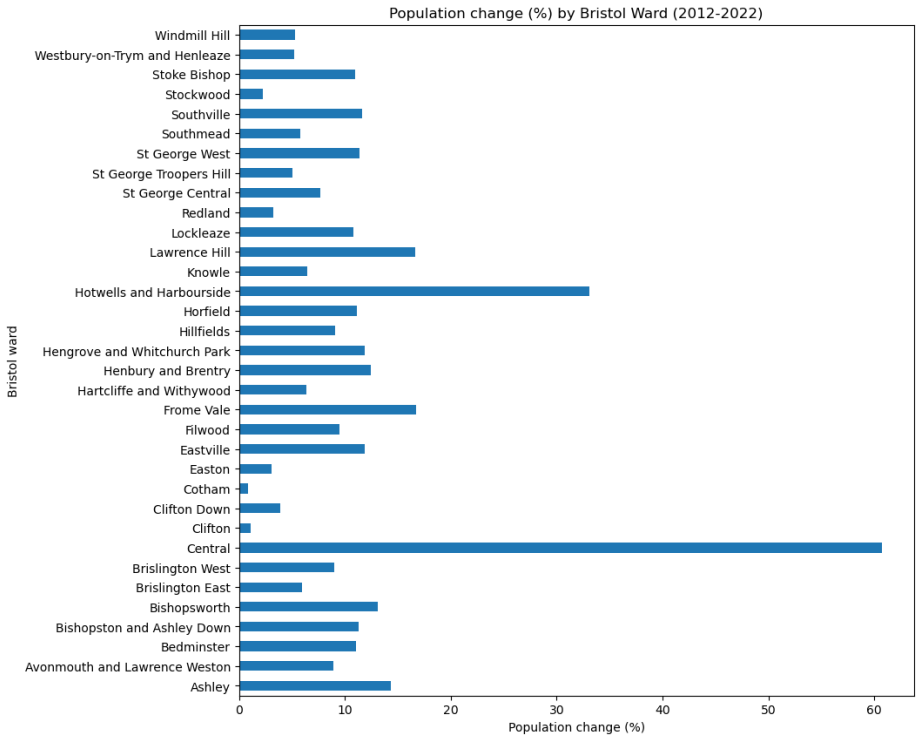


Figure C: Percentage population change by ward bar chart

Comparing population percentage increase across all wards for easily visual comparisons of change. Central is clearly seen to have experienced most change at 60.69% with Hotwells and Harbourside next at 33%. Then in contrast, a few wards have experienced little to no change at all over a decade. This was made after the bar chart to provide visual identification of the wards responsible for the very high population changes within Bristol.

Figure B: Initial population vs absolute population change scatter plot

Made a scatter plot comparing each ward population from 2012 against its absolute population change from 2012-2022. Wanted to inspect if wards with larger initial populations high a higher change of experiencing greater population increases. Calculated Pearson correlation using Pandas .corr() function to add to the analysis. The correlation value of r=0.207 shows a weak positive correlation, suggesting the hypothesis was not necessarily the case – noted that the Central ward value may have influenced this correlation value. Added in .corr() because the relationship was not particularly obvious by eye from the scatter plot.

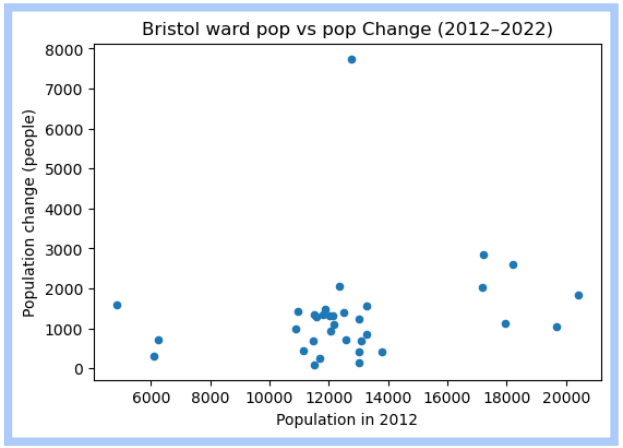


Figure 1: Choropleth insights

The choropleth was a good addition to the analysis because it presented the findings geographically and added a colour scheme for quicker visual analysis of the results.

The darker orange wards represent the greater proportional growth (especially Central and Hotwells and Harbourside) with the lighter orange shade indicating a smaller increase (most other wards).

The choropleth map is fantastic for showing the uneven population growth that was found through statistic analysis, spatially. Also, a useful map for identifying where increased population could potentially be relevant to planning decisions due to population growth etc.

Overall insights and limitations:

The main finding is that despite all of Bristol's wards experiencing population increase, this growth was uneven both statistically and spatially speaking. The histogram presented the overall distribution and outliers, the scatter plot and .corr() was for testing relationship between initial and absolute pop growth, the bar chart identified the names of the wards with the greatest proportional increases, and the choropleth showcased these results spatially across Bristol. These findings could be useful for identifying areas where further investigations into housing demand, public services and development are needed (limitation: population change itself isn't an indicator of a ward experiencing pressure but could be a useful indicator when combined with other factors e.g. more info on housing, population density, deprivation etc for a conclusion).

Lab 3 – Challenge 4

Problem/Question: Where are reported crime incidents spatially concentrated in Chicago in 2024?

- Would the geographical distribution of reported crime incidents show specific areas of concentration across Chicago in 2024?
- How DataShader could be used to visualise this large datset, especially with overlapping observations spatially?

Data source: City of Chicago Data Portal

Socrata Open Data API - Chicago Crime Dataset Documentation

- Crimes, 2001 to Present **(8+ million records):**

- Contains records of reported crime incidents including dates, crime-classifications, descriptions and locations.
- The API query used $where to select the records from 2024 with non-missing latitude longitude, then using $limit=100000 to restrict number of records returned using the API documentation as guidance.

Chose to use APIs not a downloaded dataset to make workflow more easily reproducible.

Dataset:

- **Crime data:** since it contains over 8 million records, decided to work with smaller subset of 100,000 records from 2024 like guidance from challenge instruction.
- The API response had 22 columns, so filtered down to columns id, primary_type, latitude, longitude
- For data preparation, converted the API JSON to Pandas Dataframe, used .info(), .isna().sum() to check the data
- Found that the latitude and longitude values had been returned as object types so converted to float
- Created point geometries with gpd.points_from_xy() then a GeoDataFrame with EPSG 4326
- Reprojected geometries to EPSG 3857 since basemap was the ESRI imagery basemap
- Extracted crime_x, crime_y columns (numerical) using .geometry.x, .geometry.y (from NY taxi example)

Data Challenges:

1)Large dataset + computational limitations:

- Full original dataset larger than what my laptop could process
- Initially tried a 50,000 subset to test my laptops limits then increased to 100,000 when that was fine (still filtering for 2024 and valid geometries)

2) Prepping coordinates for DataShader:

- First converted longitude and latitude from objects->floats, then had to reproject the GeoDataFrame to match ESRI basemap
- Used NY taxi example for figuring out how to extract the projected x/y coordinates from the geometries to use with hvPlot

3) Dependencies + rendering issues:

- Since DataShader needs lots of libraries like Holoviews, hvPlot, Colorcet, Bokeh nd Geoviews to run (as I found through trial and error), I used a pip install to install all of these at once
- Then experienced compatibility and rending issues due to some of the visualisation libraries, where some initial outputs worked but later didn't even after restarting kernels and reloading everything.
- Although I got the final DataShader map to work, these problems I encountered made the mapping harder than I expected.

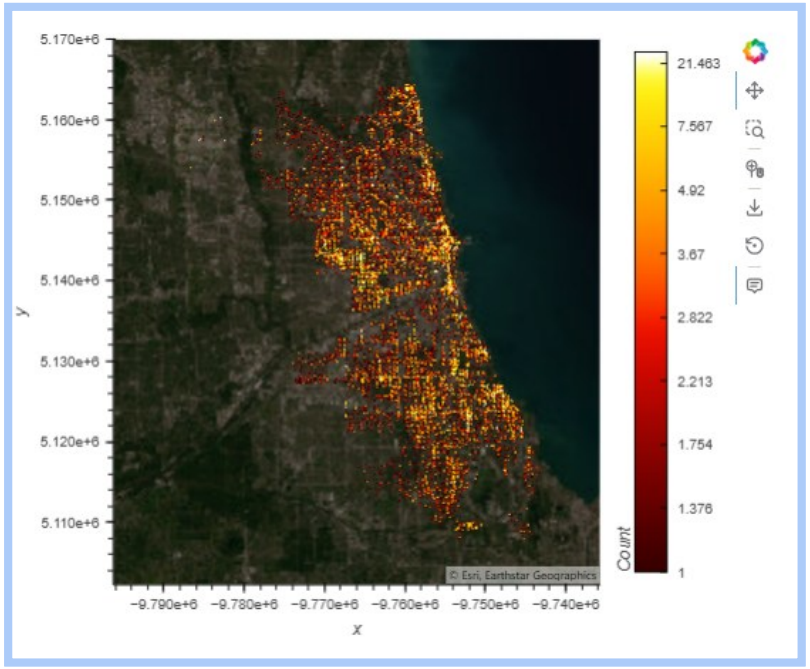


Figure 1: Spatial distribution of reported crime incidents in Chicago 2024

Used HoloViews and hvPlot with DataShader to get an interactive rasterised map of the reported locations. Used rasterize=True to aggregate overlapping points into pixels. Used cc.fire with cnorm='eq_hist' . Colour intensity represents number of observations aggregated within mapped pixels – makes it easier to visually identity areas with overlapping incidents.

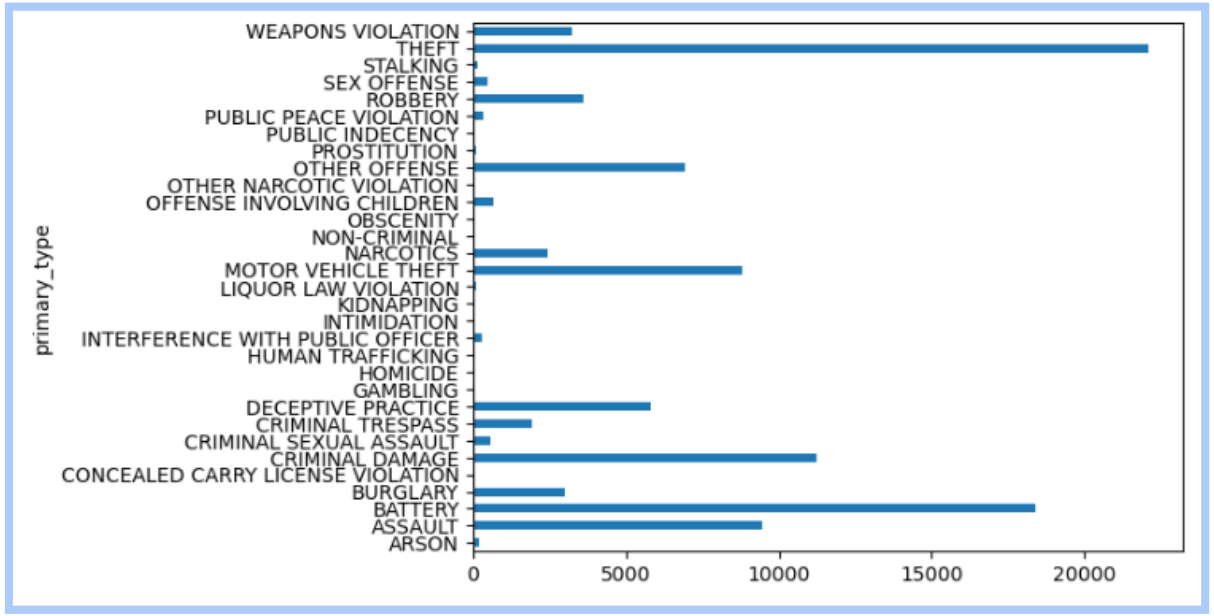
Figure 1: DataShader map insights

The map is showing the spatial distribution of the 100,000 reported crime incidents 2024 subset dataset. Rather than there being an obvious city-wide pattern, there seems to be multiple clusters of activity – where the incidents are overlapping more frequently which is producing the brighter contrasts on the map. I found that the rasterised map made the overlapping records much easier to identify and quantity in comparison with plotting each observation as an individual point on a map. However, a major limitation to accuracy with this map outcome with DataShader is that the concentrations shown are based only on the selected API records subset of 100,000 from 2024 and hence may not represent the actual distribution of all reported incidents during the whole 2024 period. The map also displays the number of records within aggregated pixels rather than came rates/seriousness of different offences hence the brighter areas do not necessarily indicate where the greatest risk of relative crime is.

Insights: Interpretations, comparisons and limitations

Figure A: Type of crime frequency chart

Decided to include the 'primary_type' column despite not being necessary for the DataShader map because I wanted to investigate the frequencies of different crime categories within the API dataset to go alongside the DataShader output – to understand if the observation overlaps has a higher probability of being due to a small number of crime types or if the crime types played more of an equal input. Used Pandas .groupby() function for this. However, the frequencies still reflect only the 100,000 subset for 2024 rather than the full 2024 dataset – so the API selection method (e.g. not random sampling) could hence affect the apparent frequency of each category. Theft has the highest recorded frequency , followed by battery, criminal damage and assault – in comparison other categories had considerably less observations. Also to note that these frequencies come from the API subset, only include recorded crime (unlikely to include all actual crime incidents), and with different offences having different reporting rates, the bar chart should not be interpreted as a reliable representation of the actual crime occurrence across Chicago across categories in 2024.



Lonboard Vs DataShader personal review

Lonboard:

- Used lonboard in challenge 3 and I found it useful for displaying the individual features interactively, with clear points shown that could then be explored by navigating round the map and zooming in and out which I enjoyed
- However, I experienced multiple issues trying to run lonboard and only managed to run it once and take quick screenshot before everything crashed and it took me a while to fix my notebook again after that.

DataShader

- Used DataShader for this Chicago dataset through hvPlot and HoloViews. Rather than displaying each crime report location individually, rasterize=True grouped observations to pixels with colour intensity indicating concentration of records. I preferred this approach as opposed to using Lonboard as for this Chicago dataset, overlapping locations would be easier to distinguish, and when zooming in.
- It was more useful to identifying spatial concentrations than lonboard since it was hard to tell the overlap quantity when just displaying points on top each other. However, when I was trying to use it, DataShader required more preparation with the same amount of library inconsistencies as lonboard, but a substantially faster runtime which I prefer.

Limitations and reliability:

1) Subsetting the original dataset:

- By not guaranteeing a random sampled subset of the full 2024 dataset this means that I cannot assume that the API 2024 subset returned a representative selection of incidents across Chicago spatially or temporally (2024). This is very important because by having a non-representative susbset, the spatial concentrations could have been exaggerated or underestimated.

2) Computer performance

- I attempted to work with much larger API responses to try to get a slightly more statistically representative subset (higher coverage or sampling method) however my laptop couldn't really handle it – so the smaller subset made the workflow more manageable but at the extent of reliable geographical conclusions that I could draw with a good enough degree of certainty. So if I were to do the analysis again, I would choose to work with a more powerful computer or instead apply a random or stratified sampling approach instead to at least have some certain representative statistical descriptions.

3) Interpretation of spatial concentrations:

- The map represents recorded incident locations rather than population-adjusted crime rates, so it doesn't currently account for population density, land use or frequency o different crim categories (hence the frequency chart I made). It would be good to introduce some additional datasets such as neighbourhood populations and boundaries to investigate if there are any areas that experienced disproportionately high levels of recorded crime. This challenge and its output also demonstrated that being able to produce a detailed map with any library does not necessarily mean that the underlying data is representative enough to support any statistically reliable geographical conclusions.